

Internet Programming

Programming and Design Assignment #1

Part 1 - Programming Exercises (5%)

Use HTML, CSS, and JavaScript to complete all the following exercises.

Exercise 1

Design a script that simulates coin tossing. Let the program toss the coin when the user clicks on a button called “**Toss**”. The user should be able to input the total number of tosses needed from a small text box. The script can count the number of times each side of the coin appears (i.e. heads or tails). Display the results of total number of heads and tails (design the display layout by yourself). The program should call a separate function `flip()` that takes no arguments and returns false for tails and true for heads.

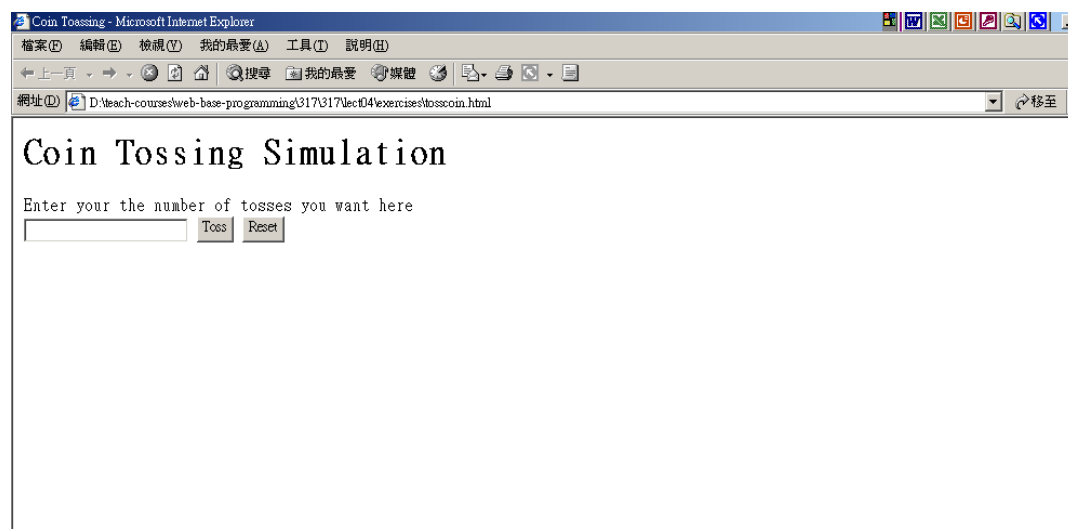
[Note: If the program realistically simulates coin tossing, each side of the coin should appear approximately 50% of total tosses].

Hints:

Create a form with a text box for input and use the following to call the `TossCoin()` function which in turn calls the `flip()` function:

```
<input name = "starttoss" type= "button" value="Toss"
onClick="TossCoin()">
```

The screen output of tossing simulation would look like the following, but you are encouraged to make a better design of the interface to make it more user friendly and show clearly the toss result summary .



Exercise 2

Computers are playing an important role in education. Write a program (in JavaScript) that will help an elementary school pupil learn division (**Dividend = Divisor x Quotient + Remainder**). Use `Math.random()` to produce two positive one-digit integer 1 to 9 as dividend and divisor (excluding 0 as small children do not have a good understanding of zero value). It should display the question such as:

Calculate 6 divide by 2?

Then the pupil types the answers into two text boxes (Quotient and Remainder) and presses the submit (My Answer) button. Your program should check the pupil's answer. If it is correct, then display "Very good!" in a pop-up window. If the answer is wrong, display "No. Please try again" in a pop-up window. You should provide pupil a button to clear the text box if he/she wants to retype the answer, and a button for generating another new question. A screen output may look like the following screen. **You are encouraged to design a better interface to make it more interesting and user-friendly for children to use.**

Calculate	6	divide by	3	?
Your Answers:				
Quotient:			Submit my answers	
Remainder			Clear	
			New Question	

Exercise 3

The use of computers in education is referred to a computer-aided instruction (CAI). One problem that developed in CAI environment is student fatigue. Varying the computer's dialog to hold the student's attention can eliminate this problem. Modify the program in Exercise 2 to give different print out of a variety of comments for both correct and incorrect answers each time.

The comment sets of responses for correct answer are as follows:

Very good!

Perfect!

Good job!

Nice work!

The comment sets of response for incorrect answers are as follows:

Incorrect, please try again.

Wrong. Try again.

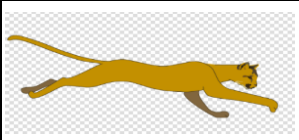
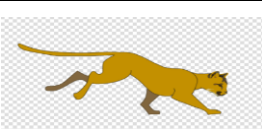
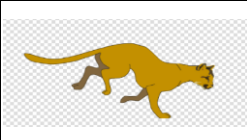
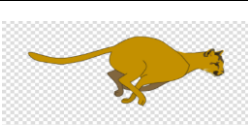
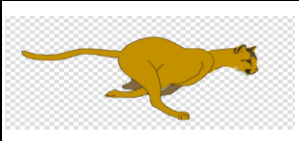
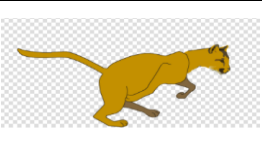
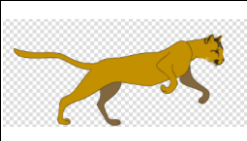
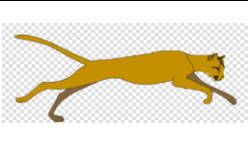
You are close!

Try harder and don't give up!

You should use random number generation to choose a number from 1 to 4 that will be used to select an appropriate response to each answer. Use an **array structure** to help the response selection.

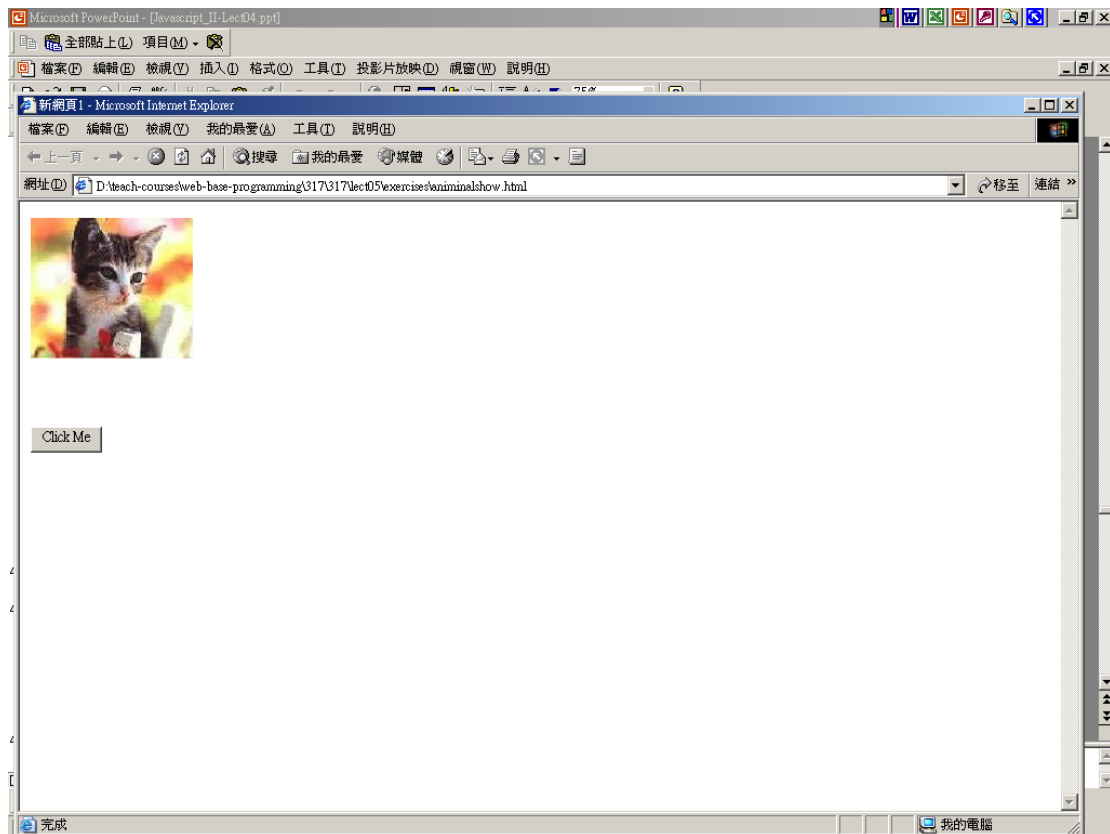
Exercise 4

Write a JavaScript program that animates a cat in motion. The images of the animation are shown are as follows:

			
cat01.png	cat02.png	cat03.png	cat04.png
			
cat05.png	cat06.png	cat07.png	cat08.png

Exercise 5

You are to create a random animal slide show, create a web page with a button when the user presses the button, a random image of an animal will be shown. Try to use an array and a random function generator to achieve your goal for creating this random animal slideshow. **Try to design a better interface to make the program more interesting and user-friendly.**



****End of Part 1****

Part 2 - Design and Build Your Personal Portfolio as a website (10%)

Introduction

In this assignment you need to develop a personal website to introduce yourself and let your lecturer or others know more about you. Use HTML, CSS, and JavaScript to create a website with a variety of **text, graphics, sound** (optional) **and animations** (moderately). The website **should** consist of *at least three pages* (students can have a maximum of **no more than 6 pages as a whole**):

Page	Physical HTML file
Home page	index.html
Personal portfolio page	portfolio.html
Hobbies page	hobbies.html

The starting (Home) page should contain a menu with **two** options; (namely personal portfolio page and hobbies) which the user can click on a link leading to a section of interest.

General Guidelines

The storyboard in the last page of this document depicts an initial plan / flow of the website (Fig 1, Fig 2, and Fig 3). The purpose of each of the three pages are listed as the following:

- index.html** - it is the starting page of the website in which are links and general introduction for the purposes of the website.
- portfolio.html** - it is the page in which a student provides some background information of himself or herself; and
- hobbies.html** - it is the page in which a student discusses his/her hobbies or things he or she is of great interest.

Specification

1. The starting page should have:

- ❑ **A header** with a title such as “Welcome to XYZ’s Home”.
- ❑ **A clear-cut page layout (with appropriate color)** that separates the header from the main page body.
- ❑ **Two hyper links or button** that allow user to click to bring the user to the appropriate page namely either the personal portfolio or hobbies page.
- ❑ **A nice and petty image as background.**
- ❑ Some background sounds or videos are optional - if you try to use them please be sure that *no copyrights are infringed or violated*.

2. The portfolio page should have:

- ❑ **Show your full name in English** with your student number.
- ❑ **Choose whatever topic** you are interested in and use this topic as theme of this page (e.g., fashion, pollutions, globalization, economics, health, education, your dreams, etc.) – **use your ideas and creativity.**
- ❑ **A good page layout with colors** that separates the heading from the body contents.
- ❑ **Some background color / image** (any color as you like) or you may use some pretty image for the same purpose.
- ❑ **A hyperlink or button** that allows the user to click and let him / her to send **an email to the owner’s registered email account** in the page.
- ❑ **Some external link to relevant websites** that supplement the topic in question.
- ❑ **A hyperlink or button** to let the user click to bring him / her immediate back to starting page (namely *index.html*).

3. The hobbies page should have:

- ❑ **Your ideas and creativity** - students need to design and complete this page on their own by applying their own creativity.
- ❑ **A good and relevant page layout** that fits the hobbies page and separates the heading and body contents.
- ❑ **Some background color** or you may use a pretty image as background for the page.

- ❑ **A description** – please provide the relevant contents (in text, graphic, images, sound, and animations) for the following **two mandatory** sections:
 - *My Hobbies*
 - *My Motto*
- ❑ **A hyperlink or button** to let the user click to bring him / her immediate back to starting page (namely *index.html*).

4. Something to note

1. Using Cascading Style Sheet (CSS) and HTML

Marking will be based on the relevant use of CSS, HTML, JavaScript. If dynamic effects are used, then they should **match** with topics/theme. Remember to put in your creativity and imagination when making these pages.

2. Using high-level tools

Use of high-level web development tool is NOT recommended. Students are expected to build the home pages with basic editing tools such as Notepad++. Students should adhere to this principle since this is a good learning opportunity for familiarizing themselves with both HTML, CSS, and JavaScript. Usually, good grades go to students sticking with this principle. Remember to make your website simple, and yet attractive and appealing is the key.

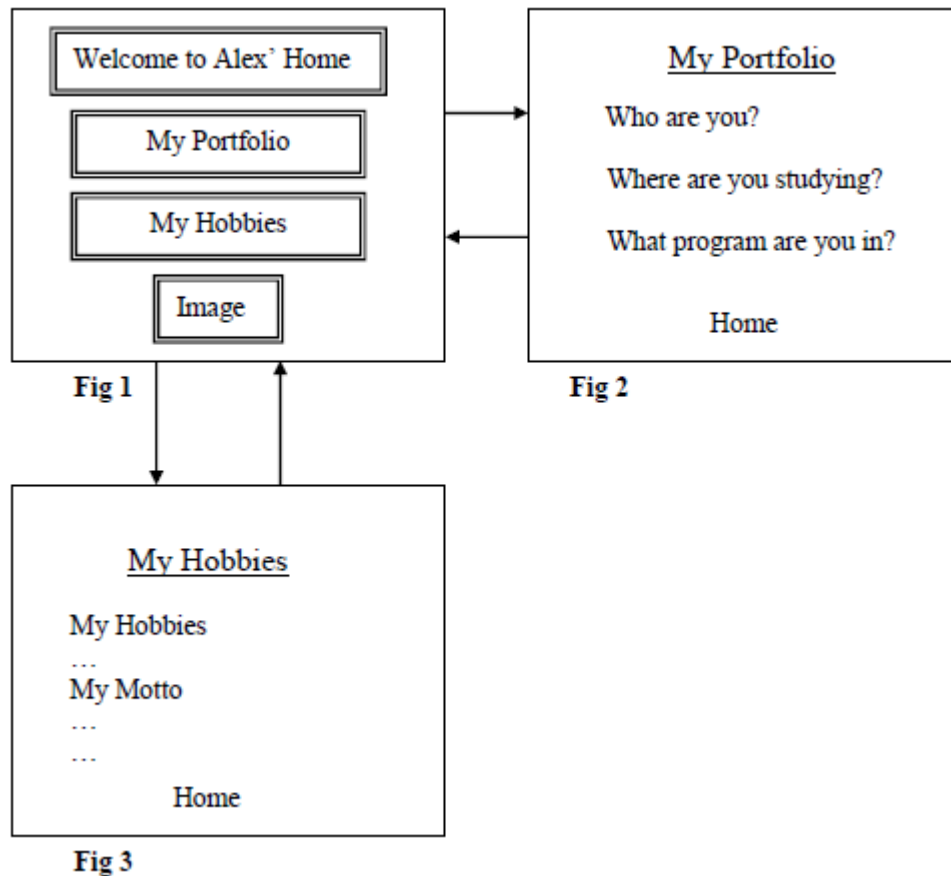
3. Using other technologies

Client-based web technologies are allowed for the creation of dynamic effects. However, only a *very small* percentage of bonus marks will be given if a student can demonstrate the appropriate use of them. The main focus of the project is still on the application of HTML, CSS, and JavaScript. So major marking is based on these core (or arguably primitive) technologies.

4. Using frame / iframe and Adding Sounds / videos

The use of *frames* / *frameset* in this project is optional if a student needs to organize information more properly. Nevertheless, using *iframe* is preferable and highly recommended whenever needed. Adding background sounds or videos are optional. They may create atmosphere / moods, but they can also be

distracting if not properly controlled and used. Make sure they are not disturbances to a user who visits your web site. Thus, it is recommended that background music should be used sparingly, and it is suggested that some control mechanism should be available to allow user to make adjustments or turn on/off the sound. The storyboard of the website is shown as follows:



****End of Part 2****